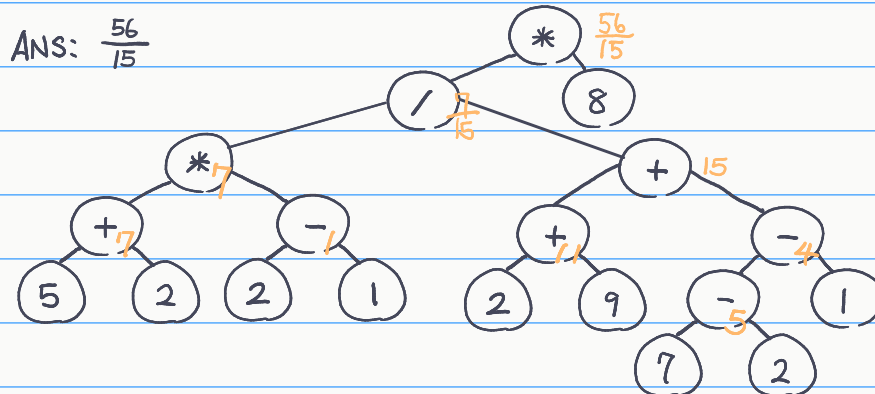


1. Describe a generalization of the Euler tour traversal to trees such that each internal node has three children. Describe how you could use this traversal to compute the height of each node in such a tree.

ANS: (1) Euler tour traversal 即在拜訪所有子節點的前後各別記錄此節點的值，最後遍訪完整棵樹後會獲得一個列表。

(2) 可以在遍歷子節點的函數中加入 depth 參數，且首次調用時 depth 為 0，接著在遍歷子節點時傳入 depth + 1，在紀錄的部份則是紀錄 (value, depth)，最後獲得一個 tuple list。欲獲得任意節點的樹高，只需找到列表中的 2 個此節點的 record a, b，並計算其之間的最大 depth，最後將此數值減去 node a 的 depth 即可得到樹高

2. Draw the binary tree representation of the following arithmetic expression:
 $((5+2)*(2-1))/((2+9)+((7-2)-1))*8$.
 and compute the value of this expression step by step (Please drawing the intermediate steps with trees).



3. A k -ary tree is a tree of which each node has at most k children. What is the maximum number of nodes in a k -ary tree of height h ? Prove your answer.

ANS: 若 k -ary 樹有最多的節點量，代表每個節點下都要掛 k 個子節點，而其總數為 $\sum_{i=0}^h k^i = \frac{k^{h+1}-1}{k-1}$ #

4. The *balance factor* of an internal node v of a proper binary tree is the difference between the heights of the right and left subtrees of v . Show how to specialize the Euler tour traversal to print the balance factors of all the internal nodes of a proper binary tree.

ANS: EULER TOUR (node) :

if not node.hasLeft() :

RETURN 0

left_height = EULERTOUR (node.left)

right_height = EULERTOUR (node.right)

bal = right_height - left_height

print (f" Balance Factor for node : { node.element } is { bal } ")

return 1 + max (left_height , right_height)